Mechanistic Estimation of Random Neural Networks

Second-round research report: aggregate response states and alternatives

Executive Summary

The strongest surviving direction is an aggregate response state, but the naive formulation T_old = sum_a sym3(G_a(x)v_a) is not economically viable if G_a is transported as a dense n x n matrix: its transport costs about 2 dense matrix multiplications per component per layer, so k=100 already costs roughly 200 units per layer. The measured M1-M3 evidence suggests that the required k is likely in the low hundreds, which would exceed the 1024-unit budget over 16 layers. Therefore the useful version must either (i) exploit additional structure in G, (ii) fit only the query-relevant state and refresh it selectively, or (iii) discover a different global quenched representation.

The most informative immediate experiment is a one-layer age-gate fit of the aggregate response state against the complete D21/D3 query family. If k below about 16-32 cannot reach 2% relative RMS, the dense-component route is effectively closed. If k around 16-32 succeeds, a selective-refresh or structured-transport variant becomes worth pursuing.

1. Q1 — Concrete fitting procedure

Let the age-gate source content be represented by
W = (A ■ P) diag(w2), so that the relevant D21 query is U = W A^T.
Use a query-weighted low-rank approximation rather than an unweighted approximation of W.

Step 1 (derived): Build a validation matrix of all four sibling query operators used by the ReLU closure: the D21 contribution, its transpose/cross arrangement, and the corresponding D3/diagonal contractions. Normalize each by its measured RMS scale.

Step 2 (derived): Solve
min_{G_a,v_a} || Q({G_a,v_a}) - Q_exact ||_F^2 / ||Q_exact||_F^2
subject to a fixed component count k,
where Q concatenates the four query families. Initialize with the rank-k SVD of W A^T (or W when A is well conditioned), then alternate least-squares updates for v_a and G_a. The fit is performed once at the age gate, not independently for every later layer.

Step 3 (conjecture): Do not fit the full tensor. Fit only the D21/D3 observables because those are the actual ReLU inputs. This avoids paying for unqueried all-distinct entries.

A useful diagnostic sequence is k = 4, 8, 16, 32, 64, 128. Record the maximum relative RMS over all four query families, not only D21. The target is <=2%.

Cost (derived): Forming W and the query matrices is O(n^2). A Lanczos/truncated-SVD-style fit is roughly O(n^2 k), about 0.001 k units at n=1024 before implementation constants. The expensive part is transport: if G is dense, G <- B G B^T costs about 2 units/component/layer, and v <- Bv is small by comparison. Thus approximately 2k units/layer, before fitting and closure overhead.

Expected k (measure-this): M1 says the raw Hadamard products are nearly full rank, while M3 says the exact contracted query has an effective rank around 230 at the 2% level. That makes k≈100-300 the plausible range for a static dense representation, which is far too expensive to transport for 16 layers. The experiment is therefore primarily a falsifier: if k<=32 works, there is a new low-dimensional phenomenon worth exploiting; if not, the straightforward aggregate-state hypothesis should be abandoned.

CPU falsifier (measure-this): On one stored .npz state at the age-4 gate, construct W, fit k=4..64, reconstruct all four query families, and compare against the exact source aggregate. A 1024x1024 SVD plus several matrix products is comfortably a sub-hour CPU experiment.

2. A more economical variant: query-space response packets

Conjecture: A better descendant of the aggregate-state idea is to store the action of the deep tensor on a fixed packet of probe vectors rather than G itself.

Choose m probe vectors r_q (for example Gaussian or deterministic Hadamard probes) and store y_q = kappa3 contracted with r_q in each required mode. Transport the probes forward/backward with the same linear map and periodically reconstruct only the D21/D3 queries.

The state size is O(nm), not O(kn^2). Its weakness is exactly the M3 result: naive random sketching had very poor variance. The new experiment should therefore use probes optimized against the measured query covariance, not iid TensorSketch features. Fit the probes by PCA/SVD of the empirical query-response covariance at the age gate.

Cost (conjecture): O(m n^2) for probe transport, approximately m/1024 dense-matmul units per layer if implemented as an n x m matrix multiplication. For m=32, this is about 0.031 unit/layer for one mode, plus reconstruction/closure. The risk is that the required m may again approach hundreds because of the flat odd signal spectrum.

3. Q2 — Other non-source formulations

Characteristic/cumulant-generating functional in a low-dimensional random-feature basis:
conjecture: unlikely under the current evidence. Parity forces the relevant realized odd component to survive conditioning on even statistics, so a basis carrying only ensemble moments cannot recover it. A random-feature implementation would inherit the observed 1/sqrt(d)-type sketching problem. First falsifier: 2, 4, 8, 16, 32, 64 feature dimensions at one age gate; require <=2% D21 RMS.

Quenched random-tensor model with covariance plus a small realized odd basis:
derived obstruction: covariance is even and cannot determine the realized odd tensor because of the exact parity result. The only potentially useful part is a compact realized residual. That reduces to the aggregate-response problem. Therefore it is not a distinct escape unless the residual is shown to have low intrinsic dimension.

Response/Jacobian statistics:
conjecture: plausible only if a response observable can be propagated without explicitly shipping the third-order tensor. A candidate is a matrix of directional derivatives of the ReLU expectation with respect to a small probe family. First falsifier: compare a 16- or

32-probe response basis with exact D21/D3 at the age gate. Failure above 2% closes this route.

Tensor-network contraction with birth index first:
derived: reordering contractions does not change the arithmetic barrier for the Hadamard query. Contracting j first gives an n x n result whose entries each aggregate n birth columns. Unless the aggregate has a new algebraic structure, this is the same O(n^3) operation that M1-M3 expose. No credible cost reduction follows from ordering alone.

TAP/cavity:
conjecture: not currently credible at this operating point. The desired object is a quenched, realized odd response, whereas conventional cavity/mean-field closures primarily compress ensemble statistics. A useful TAP-like method would need an explicit finite-n ReLU response equation and an O(n^2) update. No such closed equation is established in the provided evidence. First falsifier: derive and implement a one-layer cavity update and compare all D21 entries; >2% error closes it.

Feynman/diagrammatic resummation:
conjecture: potentially interesting theoretically, but not yet a computational escape.
The known cumulant hierarchy organizes the expansion, but the provided measurements give no evidence that the realized odd diagrams collapse to a small sufficient statistic. The first test should be whether the leading 2-3 diagram families reproduce the measured D21 residual with <=2% RMS at one layer. If not, stop.

4. Q3 — What the leaders are most likely doing

Measure-this / conjecture: the strongest evidence is not that leaders simply dropped all old sources. The closed sensitivity experiment explicitly says that removing old content produces large errors and that every layer from roughly 8 onward matters. Therefore a pure old-source truncation is inconsistent with the supplied measurements.

A more plausible single hypothesis is selective aggregation/compression of the old odd response: the leaders preserve the realized deep signal but represent it in a cheaper global state, while retaining the young-source tier. The cost coincidence (roughly 250-107 units) then reflects removal of the expensive source-by-source implementation, not removal of the underlying information.

Confirming observable: take a leader-equivalent budget implementation and measure the conditional intrinsic dimension of the aggregate old D21 response after whitening by the even covariance/transport statistics. Compute q_90, q_95, q_98 and q_99 singular dimensions for layers 8-15. If q_98 is O(10-50), a global compressed old-response state is strongly supported. If q_98 is O(200-400), the leader must be exploiting a more specialized algebraic or selective computation. If q_98 is near n, then the cost gap likely comes from a mechanism not represented by the proposed aggregate state.

5. Bottom line

Derived from the closed measurements: there is no remaining justification for another ordinary low-rank factorization of individual old legs, a covariance-only closure, a parity mean-field approximation, or a tensor-sketch surrogate. Those mechanisms are directly

constrained by the measured flat Hadamard spectra and exact parity.

Conjecture: the remaining credible class is a global quenched response representation whose state is optimized for the ReLU query family rather than for the full third-order tensor. The most promising concrete implementation is a query-space response packet with data-dependent probes, combined with occasional exact refresh at selected layers.

Measure-this: the single most valuable next CPU experiment is the age-gate intrinsic dimension test using the exact stored old-source aggregate. It directly answers whether the information itself is low-dimensional, independent of how the current V29 implementation stores it. That measurement can decisively separate 'representation problem' from 'irreducible arithmetic problem' before another large implementation is attempted.